

RESEARCH ARTICLE

Effects of Driving Background Complexity and Interface Opacity on Visual Cognition in AR-HUD Systems

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Received: 3 March 2025 | **Revised:** 25 April 2025 | **Accepted:** 30 May 2025

Keywords: AR-HUD interface design | driving background complexity | eye-tracking | information opacity | visual cognitive efficiency

ABSTRACT

The cognitive effectiveness of AR-HUD interfaces is influenced by driving background complexity (DBC) and information opacity. This study explores how they impact visual cognition and reaction efficiency using a dual-phase experimental approach. In Experiment I, a subjective evaluation classified DBC into low, medium, and high levels based on static driving scene images. This was followed by an objective assessment of the complexity of color variety, edge density, and texture features for the selected L-DBC, M-DBC, and H-DBC images. Experiment II then employed eye-tracking metrics (reaction time, mean pupil diameter, and AOI fixation duration) to evaluate participants' visual performance across 10 opacity gradients (0.1–1.0). Results revealed significant interactions between DBC and opacity levels. Under L-DBC, M-DBC, and H-DBC conditions, the relationship between information opacity and reaction times exhibited different phases. To optimize visual cognitive performance, AR-HUD opacity should be set at a minimum of 0.6. When opacity levels are below 0.7, the greater the DBC, the longer the response time for the same opacity. When the information opacity is above 0.7, quicker reaction times can be achieved, regardless of whether the DBC is high or low. These findings offer valuable design guidelines for optimizing AR-HUD text opacity in complex driving backgrounds.

1 | Introduction

The augmented reality head-up display (AR-HUD) represents a significant advancement in in-vehicle information display systems, integrating augmented reality (AR) technology with traditional head-up displays (HUDs). By superimposing digital driving assistance information onto the driver's field of vision and fusing it with the real-time driving environment, AR-HUD enables drivers to access critical information without diverting their gaze from the road [1]. This feature is particularly advantageous compared with traditional methods of information retrieval, such as using the dashboard or central control screen, as it significantly reduces the time drivers spend looking away from the road [2, 3]. Furthermore, AR-HUD enhances situational awareness by providing visualized information about

the vehicle's surroundings, including attention prompts, autonomous driving takeover performance, and driving safety categories [4]. The real-time integration of driving assistance information with the actual driving scene not only ensures an intuitive presentation but also effectively reduces the cognitive load associated with information processing [5, 6]. Tonnis et al. conducted a user study in a driving simulator, implementing an AR visualization solution for driver safety prompts. Their results demonstrated that AR-HUD can improve driving performance without increasing cognitive load. Recent studies have highlighted the significant potential of AR-HUD to enhance driving safety and efficiency.

However, during the interaction between drivers and AR-HUD, the visual cognitive effect of information within the driver's

field of vision is a critical determinant of driving safety [7–11]. This visual information comprises both the external environmental cues of the driving road and the driving assistance data displayed by the AR-HUD. The external environmental information can be influenced by factors such as road complexity, weather conditions, and traffic congestion, which in turn can interfere with the cognitive processing of AR-HUD information. Although Kim and Gabbard mentioned that the virtual shadow design in AR-HUD improves driver situation awareness about pedestrians, their findings suggest it may negatively impact awareness of other environmental elements [3]. Numerous studies have examined the impact of AR-HUD environments on driving performance and safety.

For instance, Abdi and Ben [12] investigated the use of AR-HUD under adverse weather conditions and confirmed through hypothesis verification that AR technology can enhance the driving experience and improve driving safety. Li's research [13] revealed that drivers exhibit distinct cognitive characteristics and preferences for AR-HUD interface color combinations across different driving scenarios, including daytime, nighttime, and various seasons. Currano et al. conducted a comparative investigation into driver situational awareness (SA) under AR-HUD implementation across two distinct driving scenarios [14]. Their experimental results demonstrated that while no statistically significant difference in SA levels was observed between AR-HUD and non-AR-HUD conditions, a notable inverse correlation was identified between environmental complexity and SA performance. Subsequent experimental iterations involving modulated environmental complexity revealed significant variations in SA metrics, suggesting that environmental complexity exerts a more substantial influence on driver SA than HUD interface design parameters. Additionally, nonaugmented stimuli in the environment may accidentally have a brief superimposition with AR graphics. This condition may also influence the rate of inattention blindness in turn [15]. Collectively, these findings emphasize that driving background complexity (DBC) must be treated as a critical parameter in AR-HUD interface optimization [16]. Such consideration is crucial for ensuring effective visual cognition and maintaining driving safety across various operational scenarios.

Equally important is information opacity—the transparency of AR elements—which determines whether virtual displays compete with or complement real-world visuals. Information opacity, which can be defined by the opacity and clarity of AR elements, is crucial for ensuring that AR-HUD displays do not obstruct critical environmental information or distract drivers from their primary tasks. Hou et al. [17] investigated the effects of color opacity in AR-HUD on drivers' eye movement behavior and driving performance by recording reaction times (RTs), fixation duration, and fixation counts during turns and braking. The results showed that dynamic effects significantly influence drivers' visual and cognitive demands. Graphics with T1.0 and T0.35 opacity required drivers to exert higher cognitive levels to distinguish interface information from the background environment, leading to cognitive overload. This result reveals that improper opacity settings may exacerbate cognitive load through visual competition mechanisms, thereby posing a threat to driving safety.

Hussain and Park [18] investigated the effects of real-world background complexity and color transparency levels in AR interfaces on user experience, employing these as dependent variables. Through quantitative measurements including task completion time, error rate, and visibility metrics, their study revealed significant interaction effects between these two factors. The results demonstrated that intermediate transparency levels (T0.5–T0.75) optimally balanced task performance with visual clarity.

Clearly, the opacity design of AR-HUD interfaces increases both the quantity and complexity of information in the driver's visual field, encompassing vehicle speed, navigation routes, traffic signals, obstacle alerts, and collision warnings. However, improperly designed opacity levels may induce visual clutter, impair information processing efficiency, and ultimately compromise driving performance and safety. To establish optimal opacity parameters for AR-HUD information displays across varying DBC levels, this study investigates the combined effects of these two critical factors on visual recognition performance. Through controlled manipulation of these two design parameters, we aim to establish evidence-based guidelines for optimal opacity settings in varying driving scenarios.

2 | Related Work and Research Hypothesis

2.1 | DBC and Visual Cognition

The effectiveness of AR-HUD interfaces is significantly influenced by a multitude of environmental factors and design parameters, with DBC serving as a pivotal determinant of visual cognitive performance. DBC refers to the visual intricacy of the external environment, such as road layouts, traffic density, and weather conditions, which can affect the driver's ability to process AR-HUD information efficiently [19, 20].

Existing research has established a correlation between driving context complexity and cognitive load. Through experimental investigations, Nilsson et al. [21] observed significantly elevated skin conductance (SC) and pupil diameter (PD) measurements in Hidden Exit and Intersection scenarios, indicating increased visual and cognitive demands. Wascher et al. [22] developed a series of driving scenarios stratified by cognitive load levels (low, medium, high) based on established cognitive load theory, revealing significant variations in both EEG patterns and driving performance metrics across conditions. Their results differentiated cognitive load levels across road types, demonstrating progressive increases from expressways (low) to rural roads (medium) and urban environments (high). These findings collectively suggest that excessive DBC may induce cognitive overload, potentially impairing drivers' ability to effectively integrate AR-HUD information with real-world visual cues [23].

2.2 | Information Opacity Design in AR-HUD

AR displays utilize not only the traditional red, green, and blue (RGB) color channels but also an additional Alpha (A) component, which controls the opacity of rendered content.

In AR-HUD applications, the complexity of real-world driving environments necessitates the use of semitransparent interfaces. This design choice is crucial, as opaque AR elements could obstruct critical areas of the road and further distract drivers from the primary driving task [24]. The AR-HUD projects digital information onto the real driving scene, creating a continuous superposition between “digital graphics” and “road environment events.” While this unique translucent property of AR-HUD can enrich the driving field of vision with information such as road conditions, prompts, and navigation [25], it may also compromise the visibility of this information, increase the driver’s cognitive load, and thereby affect driving safety [26].

Further studies have systematically explored the perceptual and cognitive effects of AR-HUD transparency. Notably, Falk et al. [27] demonstrated that the legibility and readability of text information displayed by AR are influenced by environmental background, opacity, and size. Complementing these findings, Ping et al. [28] thoroughly examined how shading models and opacity critically influence depth perception of virtual objects, conclusively reporting that an opacity level of 0.8 yielded the lowest depth-matching errors. In the specific context of AR-HUD displays, Hou et al. [17] meticulously investigated transparency levels in navigation graphics, revealing that a transparency setting of 0.75 effectively minimized instances of inattention blindness. Furthermore, Lopez et al. [29] comprehensively analyzed driver perception of AR-HUD opacity, ultimately concluding that a 60% transparency setting optimally balanced visibility with minimized distraction for central display elements.

These findings collectively underscore the importance of optimized transparency in AR-HUD design—where excessive opacity can dangerously obscure critical real-world objects, while insufficient opacity may substantially reduce information salience, potentially leading to visual clutter, impaired decision-making, and ultimately compromised road safety. The critically delicate balance thus required in AR-HUD transparency design must precisely ensure information clarity without overwhelming the driver or obstructing essential real-world traffic elements.

2.3 | Research Hypothesis

Optimizing opacity design in AR-HUDs across varying driving background complexities (DBC) can enhance driving performance and safety by reducing cognitive load and improving information processing efficiency. Based on the above considerations, this study aims to address the following research questions:

Do different driving background complexities (DBC) and the information opacity of AR-HUD have an impact on human visual identification efficiency?

Do different DBCs and the information opacity of AR-HUD have an impact on human visual perception characteristics?

Under different DBCs, how should the information opacity of ARUD-H be designed?

To address these research questions, this study focuses on the vehicle speed display of the AR-HUD as a starting point. It examines the oculomotor physiological responses and visual cognitive efficiency of drivers when confronted with varying degrees of opacity in AR-HUD information under different DBCs. The objective is to develop a method for text opacity design for the AR-HUD interface, thereby enhancing the usability, flexibility, and safety of AR-HUD display design.

3 | Experiment I: Evaluation of DBC

3.1 | Subjective Evaluation of DBC

3.1.1 | Participants

The experimental questionnaire was distributed to a diverse population through an online platform, ensuring broad accessibility and participation. A total of 58 valid responses were collected, comprising individuals aged 18–45 years (mean age = 28.5 years, SD = 6.2, 25 males and 33 females). Participants were categorized based on their driving frequency: 52% were occasional drivers, 21% were frequent drivers, and 26% were rare drivers. To maintain the integrity of the visual perception data, all participants were screened for color vision deficiencies, and none exhibited symptoms of color blindness or color weakness, thereby eliminating potential confounding factors related to abnormal color perception. Informed consents were obtained from all participants before they filled out the questionnaire.

3.1.2 | Materials and Methods

Referring to the methodological framework established by Li et al. [13], who employed six representative driving environment images to simulate real-world driving scenarios and assess gender-based perceptual differences in automotive HUD color scheme design, this study utilized static scene images depicting diverse driving environments to represent varying levels of DBC. Following the classification methodology proposed by Guo et al. [30], the visual complexity of these images was systematically categorized into three distinct levels (low, medium, and high) through a rigorous subjective evaluation process. A comprehensive set of 28 static scene images, which encompass diverse real-world DBC, was subjected to subjective complexity assessment using questionnaire surveys. The image database included multiple meteorological conditions (rain, sunshine, snow, and fog) and varied driving scenarios (urban areas, commercial districts, and highways), as detailed in Table 1.

The questionnaire was structured as a 5-point Likert scale, with scores ranging from 1 (*indicating minimal complexity*) to 5 (*representing maximal complexity*), allowing respondents to quantify their subjective perception of visual complexity. Participants were instructed to evaluate each image independently and assign a score corresponding to their perceived complexity level. Following the individual scoring phase, respondents were further tasked with performing a comparative ranking of all images. They identified the three driving backgrounds with the highest complexity and the three with the lowest complexity.

TABLE 1 | Samples of the DBC questionnaire.

	Sunny weather	Rainy weather	Foggy weather	Snowy weather
Complicated road conditions in urban areas	 24	 20	 26	 19
Simple road conditions in parish	 15	 6	 14	 3
Freeway	 5	 12	 21	 23
Night-time road conditions	 25	 16	 22	 11
Road condition with sidewalk ahead	 7	 28	 10	 9
Others	 17	 27	 13	 1

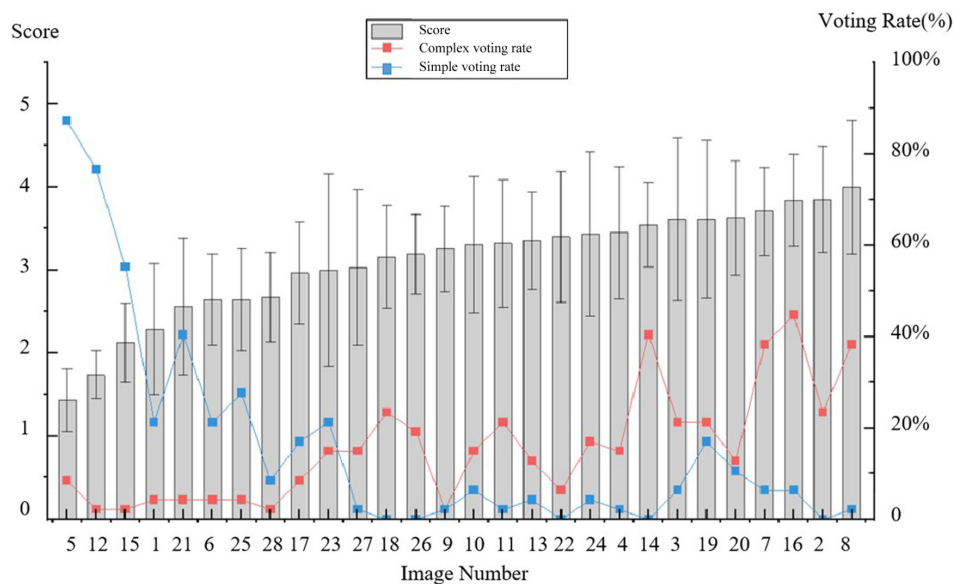


FIGURE 1 | Integrated bar and line chart representation of questionnaire survey results.

3.1.3 | Results of Data Analysis

A comprehensive reliability analysis was performed on the questionnaire data to assess the internal consistency of the collected responses. The analysis yielded a Cronbach's alpha coefficient of 0.869, significantly exceeding the established threshold of 0.8, which indicates excellent reliability for psychometric instruments. The subjective complexity scores

derived from the questionnaire are systematically presented in Figure 1.

Based on the subjective evaluation results (Figure 1), image 8 demonstrated the highest complexity level, with an average score of 4.0 (variance=0.8) and a 38% selection rate in the high-complexity ranking. Thus, it was designated as the high-driving background complexity (H-DBC). Image 5 exhibited

the lowest complexity level, with an average score of 1.43 (variance = 0.37) and an 87% selection rate in the low-complexity ranking, and was assigned as the low-driving background complexity (L-DBC). Image 9, with an average score of 3.2 (variance = 0.51) and no selections in either the high- or low-complexity rankings, was classified as the medium-driving background complexity (M-DBC). The finalized DBC images are shown in Figure 2.

3.2 | Objective Assessment of DBC

To verify the usability of the subjective evaluation results, objective indicators, including the complexity of the color variety feature [31], edge density [32], and texture features [33], were selected for verification and illustration. In this study, Python code was used to calculate each indicator.

3.2.1 | Complexity of Color Variety Feature

The complexity of color variety feature (C_s) serves as a quantitative measure of the diversity of color types within an image and can be derived from the information entropy formula. Specifically, a higher C_s value corresponds to a greater variety of color types present in the image. Zhou et al. [31] introduced a normalized C_s metric (C_s^a), which facilitates a more accurate comparison of color complexity across different images.

Using Zhou's formula for C_s^a , the C_s^a values for three DBC images (as shown in Table 2) were calculated as follows: for the L-DBC image, C_s^a was found to be 3.88; for the M-DBC image, it was 4.14; and for the H-DBC image, it reached 4.17. These results clearly demonstrate a gradient in color type complexity among the three DBC images, with the L-DBC image exhibiting relatively low color type complexity, the M-DBC image showing a higher level of complexity, and the H-DBC image possessing the highest color type complexity among the three.

3.2.2 | Edge Density

Edge density refers to the ratio of the number of edge pixels to the total number of pixels in an image. In this study, after preprocessing the image into a grayscale image, the Canny algorithm was used for edge detection, and then the edge density was calculated [32]. A higher edge density value indicates that the image contains more abundant details. The edge density value of the L-DBC image is 0.04, that of the M-DBC image is 0.07, and that of the H-DBC image is 0.08. These results show that the L-DBC image has fewer details, while the H-DBC image has richer details.

3.2.3 | Texture Features

In this study, texture features were extracted based on the gray level co-occurrence matrix (GLCM), a powerful tool for describing texture characteristics by statistically analyzing the joint probability distribution of pixel pairs in an image [33]. The following key indicators were calculated to quantify the texture properties of the datasets:

Contrast: This metric reflects the clarity and depth of texture grooves within the image. Higher values indicate deeper and more pronounced texture grooves. The Contrast values for the L-DBC, M-DBC, and H-DBC images were 0.16, 0.38, and 1.36, respectively.

Energy (ASM): This indicator represents the uniformity of the grayscale distribution in the image. A higher Energy value suggests a more uniform texture. The ASM values for the L-DBC, M-DBC, and H-DBC images were 0.29, 0.23, and 0.32, respectively.

Homogeneity (IDM): This metric reflects the local grayscale similarity within the image. Higher values indicate gentler local grayscale changes and a more homogeneous texture. The IDM values for the L-DBC, M-DBC, and H-DBC images were 0.94, 0.88, and 0.89, respectively.



FIGURE 2 | Different DBC images.

TABLE 2 | Objective assessment of DBC.

Image	C_s^a	Edge density	Texture features			
			Contrast	ASM	IDM	Correlation
L-DBC	3.88	0.04	0.16	0.29	0.94	1.00
M-DBC	4.14	0.07	0.38	0.23	0.88	0.99
H-DBC	4.17	0.08	1.36	0.32	0.89	0.95

Correlation: This measure evaluates the linear dependency between neighboring pixels, providing insight into the regularity of the texture. Higher Correlation values indicate stronger pixel-to-pixel relationships, which are often associated with more regular textures. The Correlation values for the L-DBC, M-DBC, and H-DBC images were 1.00, 0.99, and 0.95, respectively.

As seen in Table 2, taking into account the complexity of color variety feature, edge density, and texture features, it can be concluded that the L-DBC image (Figure 2) has the lowest overall complexity, the M-DBC image (Figure 2) has moderate complexity, and the H-DBC image (Figure 2) has the highest complexity.

4 | Experiment II: Effects of DBC and Information Opacity on Cognition

4.1 | Participants

The experimental sample size was determined a priori using G*Power software to ensure statistical robustness. Based on a significance level (α) of 0.05, an estimated effect size (f) of 0.25 [34], and a desired statistical power of 0.95 [35], the minimum required sample size was calculated to be 18 participants. To account for potential attrition and enhance the reliability of the findings, 23 participants (9 males, 14 females; mean age = 24 years, $SD = 2.5$) were recruited for the formal experiment. All participants met the inclusion criteria of normal or corrected-to-normal vision, confirmed absence of color vision deficiencies, and demonstrated sufficient visual acuity to perform the experimental tasks. Participants were compensated with an incentive ranging from CNY 20 to CNY 30, contingent upon their task performance accuracy. Prior to the experiment, all participants provided written informed consent after reviewing the study protocol.

4.2 | Apparatus

The experiment employed a Tobii TX300 eye-tracking system to record and analyze participants' ocular movements. The device operated at a sampling rate of 300 Hz. The video stimuli were displayed on a monitor with a resolution of 1280×1024 pixels, a luminance of 300 cd/m^2 , and a refresh rate of 144 Hz. The eye tracker was synchronized with the monitor during recording. To maintain consistent lighting conditions throughout the experiments, all trials were conducted in an enclosed laboratory with fixed lighting parameters (ambient light at approximately 550 lx), to ensure full reproducibility of the experiment.

4.3 | Experimental Materials

4.3.1 | Text Opacity Design of AR-HUD Vehicle Speed Display

Extensive research investigations have demonstrated that a significant proportion of vehicular accidents result from drivers' inaccurate speed perception [36, 37]. Vehicle speed monitoring is a critical component of driving performance, requiring continuous attention and precise interpretation by drivers. A well-designed AR-HUD vehicle speed display has

the potential to mitigate accident rates by enhancing speed perception accuracy. Speed-related information typically includes "current speed" and "speed limit." Through a comprehensive analysis of existing AR-HUD implementations in commercial vehicles, it was identified that the "speed limit" information predominantly adopts a standardized design, characterized by a red circular border with black numerical font (see Table 3). Consequently, this design scheme was implemented for the "speed limit" information in our experimental setup. For the "current speed" information, market analysis revealed a prevalent use of white numerical displays. Building on Zhou et al.'s findings [38], which indicate that white, cyan, and green are optimal for automotive AR-HUD interfaces, a white DS-Digital font at 100-pt size was utilized for the "current speed" information. The complete AR-HUD speed display design was created using Adobe Photoshop 2023 software, as shown in Figure 3.

4.3.2 | Experimental Samples

The experimental design incorporated two independent variables: DBC and vehicle speed text opacity. The DBC was operationalized at three distinct levels, corresponding to the L-DBC, M-DBC, and H-DBC image depicted in Figure 2. Vehicle speed text opacity was implemented across 10 incremental levels (0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, and 1.0), where the number 0 indicates complete transparency and 1 indicates complete opacity. Experiment II included a total of 30 samples.

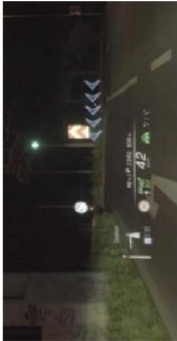
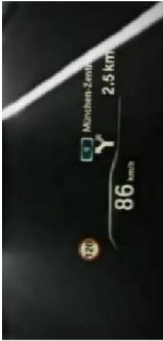
In this study, a total of 30 experimental video materials were produced using Adobe Premiere Pro 2020 software. Each video maintained a constant level of DBC and a static "speed limit" sign, while the "current speed" value dynamically increased over time. Specifically, the "current speed" started below the "speed limit" and continuously rose to exceed it. The duration of each video ranged from 5 to 12 s, with the "current speed" surpassing the "speed limit" at a variable time point after 3 s into the video.

4.4 | Experimental Procedure

Before Experiment II starts, participants were positioned such that their visual axis was aligned parallel to the center of the display screen at a viewing distance of 50–60 cm. The experimental protocol commenced with a five-point calibration procedure for the eye-tracking system. Participants were then presented with detailed instructions, emphasizing the requirement to promptly click the mouse when the "current speed" value matched the "speed limit" threshold. The experimental sequence consisted of three practice trials followed by 30 formal trials. Before the formal experiment, participants were required to read the instructions and complete a simulated trial using three experimental video samples to familiarize themselves with the procedures.

As illustrated in Figure 4, the formal trials were organized into three groups, with each group containing 10 randomly ordered video stimuli. Each trial followed a standardized sequence: (1) presentation of a fixation cross ("+") for 1000 ms, (2) a blank screen interval of 1000 ms, and (3) random presentation of the

TABLE 3 | Comparative analysis of current AR-HUD interface design.

Brand name	AR-HUD interface	“Speed limit”		“Current speed”	
		Display format	Display position	Display format	Display position
Mercedes-Benz S-Class		Standard traffic signs	Placed to the left of the display area alongside road signs	White numeric image	Center of the display area
Volkswagen ID.4		Standard traffic signs	Placed at the center of the display area along with the current speed	White numeric image	Center of the display area
Audi Q4 e-tron		Standard traffic signs	Placed independently at the far right of the display area	White numeric image	Left side of the display area
BMW		Standard traffic signs	Placed independently at the far left of the display area	White numeric image	Center of the display area
Huawei Aito M7		Transparent background Standard traffic signs	Placed on the left side of the display area alongside the current speed	White numeric image	Left side of the display area

(Continues)

TABLE 3 | (Continued)

Brand name	AR-HUD interface	“Speed limit”		“Current speed”	
		Display format	Display position	Display format	Display position
Hongqi E-HS9		Standard traffic signs	Placed at the center of the display area along with the current speed	White numeric image	Center of the display area
Feifan R7		Transparent background Standard traffic signs	Placed on the left side of the display area along with other traffic signs	White numeric image	Center of the display area



FIGURE 3 | Schematic representation of vehicle speed display interface design (opacity = 1.0).

video stimulus. Participants were instructed to click the mouse immediately upon detecting the “current speed” reaching the “speed limit” value, at which point the trial terminated and the subsequent trial commenced. Following each group of 10 trials, participants were provided with a 3-min rest period to mitigate visual fatigue. The experimental session concluded upon completion of all three trial groups.

Throughout the experimental trials, participants’ eye-movement behavior and RTs were systematically recorded using the eye-tracking system, to quantitatively assess visual perception characteristics and identification efficiency. As shown in Figure 5, three distinct areas of interest (AOIs) were defined for each video stimulus: the current speed AOI, the speed limit AOI, and the background AOI.

5 | Results and Analysis of Experiment II

5.1 | Data Preprocessing

The experimental protocol yielded eye-tracking data from all 23 participants. To investigate identification efficiency and visual perceptual characteristics, three key metrics were analyzed: (1) RT, defined as the interval between the target speed threshold appearance and the participant’s response click; (2) gaze duration in the background AOI (GDB-AOI); and (3) average pupil diameter (APD). The APD provides clearly feedback for evaluating the cognitive load during driving. Čegovnik et al. [39] proposed that pupillometry can be used to assess changes in drivers’ cognitive load. Eye trackers accurately identified that pupil diameter increases with the difficulty of secondary tasks. Shi et al. [40] pointed out that pupil diameter can generally reflect variations in mental load across different road environments. Significant increases in average pupil size were observed after performing driving tasks of varying difficulties. RT served as a critical indicator of response accuracy, with negative values indicating premature responses (target-not-present errors) and sub-200-ms responses suggesting anticipatory guessing behavior rather than genuine visual processing [41]. Following established cognitive response thresholds, data points exhibiting these characteristics were excluded as invalid responses. After rigorous data validation, 663 valid data points were retained for subsequent analysis.

5.2 | Normality Assessment

As seen in Table 4, the normality of the experimental data, including RT, APD, and GDB-AOI, was assessed using the

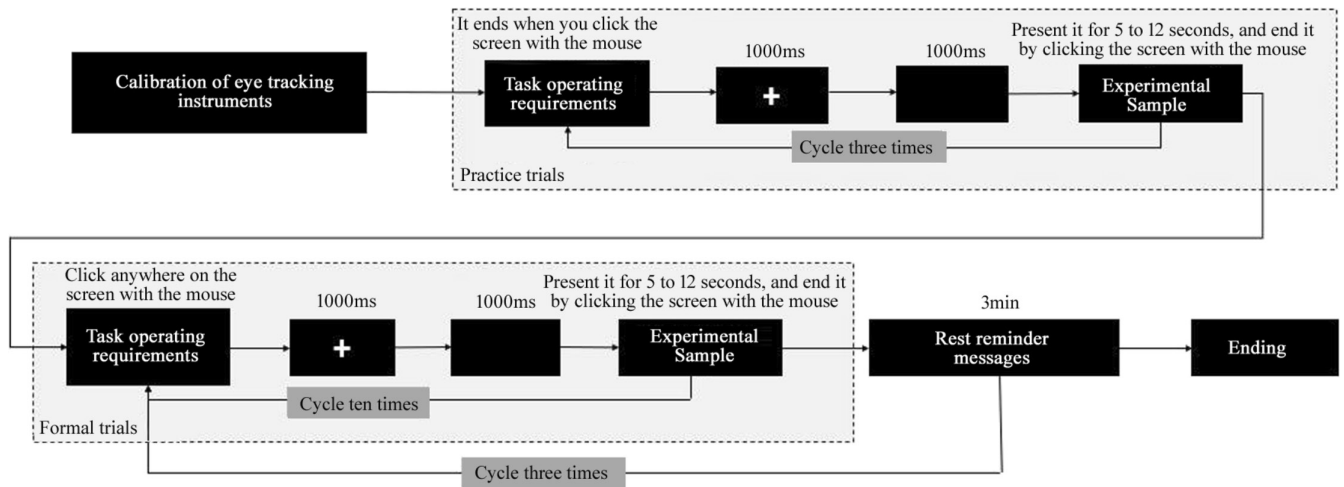


FIGURE 4 | Flow chart of Experiment II.



FIGURE 5 | Defining areas of interest (AOIs) for experimental stimuli.

Kolmogorov–Smirnov test. The results indicated that GDB-AOI exhibited nonnormal distribution across all DBC and text opacity ($p < 0.05$), while APD demonstrated normal distribution under all experimental ($p = 0.2 > 0.05$). RT data showed mixed distribution patterns: under varying text opacity levels, some data displayed nonnormal distribution ($p = 0.004, 0.045, 0.027 < 0.05$), and all DBC resulted in nonnormally distributed RT data ($p < 0.05$).

5.3 | Analysis of Behavioral Data

To analyze the nonnormally distributed RT data, a generalized linear model (GLM) was implemented to examine the effects of DBC and text opacity design on RT. The GLM effect tests revealed statistically significant main effects for both DBC ($\chi^2 = 86.809, p < 0.05$) and text opacity ($\chi^2 = 194.794, p < 0.05$). Furthermore, a significant interaction effect was observed between DBC and text opacity ($\chi^2 = 67.605, p < 0.05$).

5.3.1 | Effect of Text Opacity on RTs

Under L-DBC, the analysis of RTs across varying text opacity levels revealed distinct patterns (Table 5). Specifically, among the 0.1, 0.2, and 0.3 text opacity levels, no statistically significant differences in RTs were observed between any pair ($p > 0.05$). Similarly, no statistically significant differences in RTs were found between any pair of text opacity levels (0.7, 0.8, 0.9, and

1.0) ($p > 0.05$). These findings suggest threshold effects in the relationship between text opacity and reaction performance under L-DBC.

Under M-DBC, the analysis of RTs across varying text opacity levels demonstrated specific patterns (Table 6). Notably, among the 0.1, 0.2, and 0.3 text opacity levels, no statistically significant differences in RTs were observed between any pair ($p > 0.05$). Similarly, RTs showed no significant variations across opacity levels of 0.6, 0.7, 0.8, 0.9, and 1.0 ($p > 0.05$). These findings suggest threshold effects in the relationship between text opacity and reaction performance under M-DBC.

Under H-DBC, the analysis of RTs across varying text opacity levels revealed distinct patterns (Table 7). Specifically, among the 0.1, 0.2, 0.3, 0.4, 0.5, and 0.6 text opacity levels, no statistically significant differences in RTs were observed between any pair ($p > 0.05$). Among the 0.7, 0.8, 0.9, and 1.0 text opacity levels, no statistically significant differences in RTs were observed between any pair too ($p > 0.05$). These results indicate consistent performance thresholds in driver response behavior under H-DBC.

5.3.2 | Effect of DBC on RTs

As seen in Table 8, pairwise comparisons of RTs across different DBCs revealed significant patterns, as detailed in Table 8. At lower text opacity levels (0.1 and 0.2), RTs differed significantly among

TABLE 4 | Kolmogorov–Smirnov test results for DBC and text opacity.

		RT (s)		APD (mm)		GDB—AOI	
		<i>Z</i>	<i>p</i>	<i>Z</i>	<i>p</i>	<i>Z</i>	<i>p</i>
Text opacity	0.1	0.100	0.178	0.066	0.200*	0.208	<0.001
	0.2	0.073	0.200*	0.045	0.200*	0.220	<0.001
	0.3	0.110	0.055	0.068	0.200*	0.196	<0.001
	0.4	0.088	0.200*	0.087	0.200*	0.234	<0.001
	0.5	0.114	0.027	0.061	0.200*	0.215	<0.001
	0.6	0.110	0.045	0.086	0.200*	0.226	<0.001
	0.7	0.077	0.200*	0.072	0.200*	0.213	<0.001
	0.8	0.082	0.200*	0.047	0.200*	0.281	<0.001
	0.9	0.106	0.075	0.068	0.200*	0.178	<0.001
	1.0	0.057	0.200*	0.069	0.200*	0.155	<0.001
DBC	L-DBC	0.088	<0.001	0.042	0.200*	0.234	<0.001
	M-DBC	0.076	0.004	0.044	0.200*	0.192	<0.001
	H-DBC	0.076	0.004	0.036	0.200*	0.179	<0.001

*The significance level is 0.05.

TABLE 5 | Pairwise comparisons of RTs at different transparencies under L-DBC.

		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0.1	<i>MD</i>	—	0.048	0.016	0.075*	0.094*	0.008	0.133*	0.095*	0.095*	0.102*
	<i>p</i>	—	0.116	0.599	0.014	0.002	0.801	0.000	0.002	0.003	0.001
0.2	<i>MD</i>		—	−0.032	0.027	0.046	−0.041	0.084*	0.047	0.047	0.054
	<i>p</i>		—	0.295	0.380	0.132	0.192	0.006	0.128	0.141	0.077
0.3	<i>MD</i>			—	0.059	0.078*	−0.008	0.117*	0.078*	0.079*	0.086*
	<i>p</i>			—	0.054	0.011	0.789	0.000	0.010	0.013	0.005
0.4	<i>MD</i>				—	0.019	−0.067*	0.058	0.020	0.020	0.027
	<i>p</i>				—	0.529	0.030	0.059	0.520	0.531	0.373
0.5	<i>MD</i>					—	−0.086*	0.039	0.000	0.001	0.008
	<i>p</i>					—	0.005	0.209	0.988	0.985	0.794
0.6	<i>MD</i>						—	0.125*	0.087*	0.087*	0.094*
	<i>p</i>						—	0.000	0.005	0.007	0.002
0.7	<i>MD</i>							—	−0.038	−0.038	−0.031
	<i>p</i>							—	0.214	0.233	0.319
0.8	<i>MD</i>								—	0.000	0.008
	<i>p</i>								—	0.996	0.806
0.9	<i>MD</i>									—	0.007
	<i>p</i>									—	0.816
1.0	<i>MD</i>										—
	<i>p</i>										—

*The significance level is 0.05.

TABLE 6 | Pairwise comparisons of RTs at different transparencies under M-DBC.

		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0.1	MD	—	−0.004	0.053	0.068*	0.111*	0.180*	0.182*	0.197*	0.239*	0.214*
	<i>p</i>	—	0.914	0.110	0.034	0.000	0.000	0.000	0.000	0.000	0.000
0.2	MD		—	0.057	0.071*	0.115*	0.184*	0.185*	0.200*	0.242*	0.218*
	<i>p</i>		—	0.085	0.024	0.000	0.000	0.000	0.000	0.000	0.000
0.3	MD			—	0.015	0.058	0.127*	0.128*	0.143*	0.185*	0.161*
	<i>p</i>			—	0.645	0.070	0.000	0.000	0.000	0.000	0.000
0.4	MD				—	0.043	0.112*	0.113*	0.128*	0.170*	0.146*
	<i>p</i>				—	0.162	0.000	0.000	0.000	0.000	0.000
0.5	MD					—	0.068*	0.070*	0.085*	0.127*	0.102*
	<i>p</i>					—	0.025	0.022	0.005	0.000	0.001
0.6	MD						—	0.002	0.017	0.059	0.034
	<i>p</i>						—	0.962	0.588	0.059	0.269
0.7	MD							—	0.015	0.057	0.033
	<i>p</i>							—	0.622	0.066	0.290
0.8	MD								—	0.042	0.017
	<i>p</i>								—	0.177	0.573
0.9	MD									—	−0.025
	<i>p</i>									—	0.428
1.0	MD										—
	<i>p</i>										—

*The significance level is 0.05.

L-DBC, M-DBC, and H-DBC ($p < 0.05$), with the shortest RTs under L-DBC and the longest RTs under H-DBC. For intermediate opacity levels (0.3, 0.4, and 0.5), significant differences were observed between L-DBC and M-DBC ($p < 0.05$), as well as between L-DBC and H-DBC ($p < 0.05$). No significant difference was observed between M-DBC and H-DBC ($p > 0.05$). At 0.6 opacity, significant differences were detected between H-DBC and L-DBC ($p < 0.05$), as well as between H-DBC and M-DBC ($p < 0.05$). RTs under high complexity were significantly longer than those under low and medium complexities. However, at the higher opacity levels (0.7, 0.8, 0.9, and 1.0), no significant differences in RTs were observed across any complexity conditions ($p > 0.05$).

5.4 | Analysis of Eye-Movement Data

5.4.1 | APD

Given that the APD data were normally distributed ($p = 0.200 > 0.05$), a general linear model was employed to examine the effects of DBC and text opacity on APD. The text opacity did not significantly influence APD ($F = 0.231$, $p = 0.990 > 0.05$). Similarly, the interaction effect between DBC and text opacity was not significant ($F = 0.196$, $p = 1.000 > 0.05$). However, DBC significantly affected the APD ($F = 17.592$, $p < 0.001$).

To further investigate the differences in APDs across varying levels of DBC, an LSD post hoc test was performed. As shown in Table 9, the results revealed significant differences in APDs between H-DBC and the other two conditions (L-DBC and M-DBC; $p < 0.001$). Specifically, APDs under H-DBC conditions were significantly larger than those under both L-DBC and M-DBC. In contrast, no significant difference was observed between L-DBC and M-DBC ($p = 0.108 > 0.05$).

5.4.2 | GDB-AOI

Given that the GDB-AOI exhibited a nonnormal distribution ($p < 0.05$), a generalized linear model (GLM) was employed to assess the effects of DBC and text opacity on GDB-AOI. The GLM effect tests revealed that text opacity did not significantly influence GDB-AOI ($\chi^2 = 15.697$, $p = 0.073 > 0.05$). Similarly, the interaction effect between DBC and text opacity was not statistically significant ($\chi^2 = 25.011$, $p = 0.125 > 0.05$). However, DBC demonstrated a significant effect on GDB-AOI ($\chi^2 = 59.476$, $p < 0.001$).

To further explore the differences in GDB-AOI across varying levels of DBC, Tamhane's post hoc test was conducted. As presented in Table 10, the results indicated significant differences in GDB-AOI between L-DBC and the other two conditions (M-DBC and H-DBC; $p < 0.001$). Specifically, the GDB-AOI under L-DBC

TABLE 7 | Pairwise comparisons of RTs at different transparencies under H-DBC.

		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0.1	<i>MD</i>	—	−0.023	−0.029	−0.020	−0.016	−0.046	0.098*	0.105*	0.115*	0.082*
	<i>p</i>	—	0.470	0.375	0.522	0.607	0.152	0.002	0.001	0.000	0.010
0.2	<i>MD</i>		—	−0.006	0.003	0.007	−0.023	0.121*	0.127*	0.138*	0.105*
	<i>p</i>		—	0.862	0.927	0.827	0.467	0.000	0.000	0.000	0.001
0.3	<i>MD</i>			—	0.008	0.012	−0.018	0.127*	0.130*	0.144*	0.110*
	<i>p</i>			—	0.791	0.695	0.584	0.000	0.000	0.000	0.001
0.4	<i>MD</i>				—	0.004	−0.026	0.118*	0.124*	0.135*	0.102*
	<i>p</i>				—	0.898	0.409	0.000	0.000	0.000	0.001
0.5	<i>MD</i>					—	−0.030	0.114*	0.120*	0.132*	0.098*
	<i>p</i>					—	0.341	0.000	0.000	0.000	0.002
0.6	<i>MD</i>						—	0.144*	0.150*	0.161*	0.128*
	<i>p</i>						—	0.000	0.000	0.000	0.000
0.7	<i>MD</i>							—	0.006	0.017	−0.016
	<i>p</i>							—	0.848	0.584	0.601
0.8	<i>MD</i>								—	0.011	−0.022
	<i>p</i>								—	0.723	0.481
0.9	<i>MD</i>									—	−0.034
	<i>p</i>									—	0.295
1.0	<i>MD</i>										—
	<i>p</i>										—

*The significance level is 0.05.

was significantly shorter than those under M-DBC and H-DBC conditions. In contrast, no significant difference was observed between M-DBC and H-DBC conditions ($p = 0.219 > 0.05$).

6 | Discussion

6.1 | Text Opacity Under Different DBCs

Based on Experiment II's findings, the complexity of the driving background, the AR-HUD opacity, and their interaction all exert significant influences on RTs. As shown in Figure 6, increased DBC is associated with longer RTs for visual identification. As AR-HUD opacity increases from low (0.1) to high (1.0) levels, RTs initially remain consistently high, then exhibit a significant decline, and eventually stabilize at a relatively low level.

Further analysis of the data, as shown in Figure 6, reveals distinct patterns under varying conditions. Under L-DBC, RTs are generally prolonged when text transparency is below 0.3, whereas transparency levels above 0.7 result in shorter RTs. Specifically, under L-DBC, visual identification performance improves when text transparency exceeds 0.7, yielding relatively shorter RTs. Similarly, under M-DBC, RTs are extended when text transparency is below 0.3, but transparency levels above

0.6 lead to shorter RTs. Under these conditions, visual identification performance is optimized, and shorter RTs are achieved when text transparency exceeds 0.6. Under H-DBC, RTs are generally prolonged when text transparency is below 0.6, while transparency levels above 0.7 result in shorter RTs. Under these conditions, visual identification performance is enhanced, and shorter RTs are achieved when text transparency exceeds 0.7.

In conclusion, to optimize visual identification performance, AR-HUD transparency should be set at a minimum of 0.6. When AR-HUD transparency is set at 0.7, visual identification efficiency remains relatively stable, and faster RTs are achieved, regardless of whether the DBC is high or low. These findings provide valuable insights for the design of AR-HUD systems that enhance driver performance under varying environmental conditions.

6.2 | Influence of DBC on Visual Perception Characteristics

Based on the analysis of physiological indicators of eye movement, the AR-HUD opacity design demonstrates no significant influence on either the APD or GDB-AOI. Similarly, the interaction effect between text opacity and DBC does not significantly influence these metrics. In contrast, DBC exhibits a significant

TABLE 8 | Pairwise comparisons of RTs under different DBCs.

Text opacity	Comparison between complexity group (Group 1—Group 2)	Mean difference (I—J)	Standard error	p	95% CI	
					Lower limit	Top line
0.1	L-DBC—M-DBC	−0.1384*	0.0318	0.0000	−0.2007	−0.0760
	L-DBC—H-DBC	−0.0381*	0.0314	0.0250	−0.0997	0.0234
	M-DBC—H-DBC	0.1003*	0.0325	0.0020	0.0366	0.1640
0.2	L-DBC—M-DBC	−0.1902*	0.0314	0.0000	−0.2517	−0.1286
	L-DBC—H-DBC	−0.1094*	0.0310	0.0000	−0.1702	−0.0486
	M-DBC—H-DBC	0.0808*	0.0317	0.0110	0.0186	0.1430
0.3	L-DBC—M-DBC	−0.1012*	0.0323	0.0020	−0.1645	−0.0380
	L-DBC—H-DBC	−0.0827*	0.0314	0.0080	−0.1443	−0.0212
	M-DBC—H-DBC	0.0185	0.0329	0.5740	−0.0460	0.0831
0.4	L-DBC—M-DBC	−0.1453*	0.0310	0.0000	−0.2061	−0.0845
	L-DBC—H-DBC	−0.1334*	0.0307	0.0000	−0.1936	−0.0733
	M-DBC—H-DBC	0.0310	0.0310	0.7030	−0.0490	0.0726
0.5	L-DBC—M-DBC	−0.1212*	0.0307	0.0000	−0.1814	−0.0611
	L-DBC—H-DBC	−0.1488*	0.0307	0.0000	−0.2090	−0.0887
	M-DBC—H-DBC	−0.0276	0.0307	0.3680	−0.0877	0.0325
0.6	L-DBC—M-DBC	0.0344	0.0310	0.2670	−0.0264	0.0952
	L-DBC—H-DBC	−0.0919*	0.0317	0.0040	−0.1542	−0.0298
	M-DBC—H-DBC	−0.1264*	0.0314	0.0000	−0.1880	−0.0649
0.7	L-DBC—M-DBC	−0.0893	0.0307	0.0640	−0.1495	−0.0293
	L-DBC—H-DBC	−0.0725	0.0307	0.0780	−0.1327	−0.0125
	M-DBC—H-DBC	0.0168	0.0307	0.5840	−0.0433	0.0769
0.8	L-DBC—M-DBC	−0.0361	0.0307	0.2390	−0.0962	0.0240
	L-DBC—H-DBC	−0.0285	0.0310	0.3580	−0.0893	0.0323
	M-DBC—H-DBC	0.0076	0.0310	0.8070	−0.0532	0.0684
0.9	L-DBC—M-DBC	0.0056	0.0321	0.8610	−0.0574	0.0686
	L-DBC—H-DBC	−0.0174	0.0325	0.5920	−0.0811	0.0463
	M-DBC—H-DBC	−0.0230	0.0317	0.4680	−0.0852	0.0392
1.0	L-DBC—M-DBC	−0.0264	0.0307	0.3900	−0.0865	0.0338
	L-DBC—H-DBC	−0.0584	0.0314	0.0630	−0.1200	0.0031
	M-DBC—H-DBC	−0.0321	0.0314	0.3070	−0.0936	0.0295

*The significance level is 0.05.

impact on both the APD and AOI gaze duration. The APD serves as a critical indicator for assessing driver cognitive load [16,42], while GDB-AOI reflects the allocation of visual attention [43, 44]. Given these findings, it can be inferred that DBC plays a pivotal role in modulating cognitive load and attention distribution in the context of AR-HUD usage. These results highlight the importance of considering environmental complexity in the design and evaluation of AR-HUD systems to optimize driver performance and safety.

6.2.1 | Influence of DBC on Cognitive Load

The APD under H-DBC is significantly larger than that under L-BDC and M-BDC. Given that cognitive load is associated with changes in APD [45, 46], it can be inferred that H-BDC induces a higher cognitive load. In this experiment, the H-BDC was represented by a nighttime environment, while L-BDC and M-BDC were depicted by daytime environments. No significant difference in APD was observed between L-BDC and M-BDC.

TABLE 9 | LSD comparison of the average pupil diameters under different DBCs.

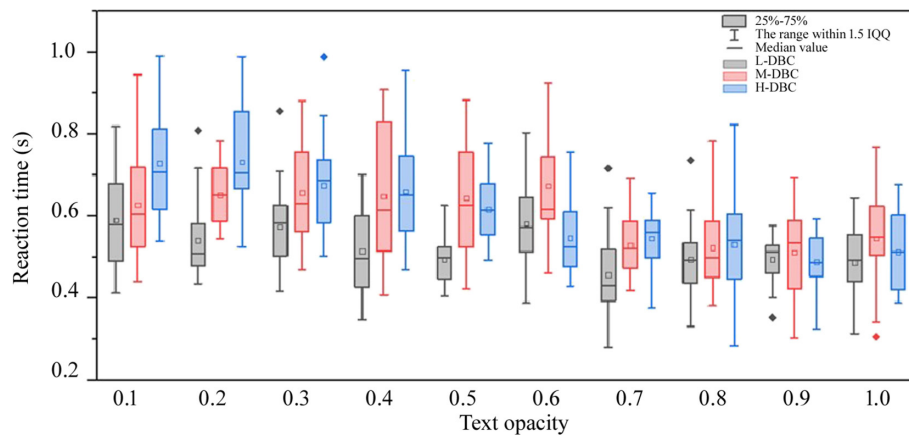
(I) complexity group	(J) complexity group	Mean difference (I – J)	Standard error	p	95% CI	
					Lower limit	Top line
L-DBC	M-DBC	–0.0848	0.0527	0.1080	–0.1883	0.0188
	H-DBC	–0.3116*	0.0528	0.0000	–0.4153	–0.2080
M-DBC	L-DBC	0.0848	0.0527	0.1080	–0.0188	0.1883
	H-DBC	–0.2269*	0.0532	0.0000	–0.3314	–0.1224
H-DBC	L-DBC	0.3116*	0.0528	0.0000	0.2080	0.4153
	M-DBC	0.2269*	0.0532	0.0000	0.1224	0.3314

*The significance level is 0.05.

TABLE 10 | Multiple comparisons of GDB-AOI under different DBCs.

(I) complexity group	(J) complexity group	Mean difference (I – J)	Standard error	p	95% CI	
					Lower limit	Top line
L-DBC	M-DBC	–5.609%*	1.018%	0.000	–8.053%	–3.165%
	H-DBC	–7.920%*	1.097%	0.000	–10.554%	–5.287%
M-DBC	L-DBC	5.609%*	1.018%	0.000	3.165%	8.053%
	H-DBC	–2.311%	1.313%	0.219	–5.461%	0.838%
H-DBC	L-DBC	7.920%*	1.097%	0.000	5.287%	10.554%
	M-DBC	2.311%	1.313%	0.219	–0.838%	5.461%

*The significance level is 0.05.

**FIGURE 6** | Box plot of reaction time across conditions.

Furthermore, nighttime driving is widely recognized as one of the most complex driving conditions due to the restricted availability of visual information and the inherent limitations of the human visual system [47]. Yang et al. [48] conducted a study on the visual functions and driving performance of older adults under daytime, nighttime, and nighttime glare conditions. Driving performance under nighttime conditions is significantly worse than under daytime conditions, and the participants' vision declines in nighttime conditions. This finding suggests that the significant increase in APD observed in high-display background complexity (H-DBC) scenarios may stem from a

dual-mechanism: It includes both the physiological pupil dilation (pupillary light reflex) triggered by low-light environments and the neural regulation caused by increased cognitive resource demands. Although the experiment controlled the output of the display device through a constant screen brightness (300 cd/m²), the dark content of the nighttime scenarios themselves might still elicit pupil responses independent of cognitive load. But Peysakhovich et al. [49] found that the impact of working memory load on the tonic pupil diameter was modulated by the level of luminance, the increase in tonic pupil diameter with the load being larger under lower luminance. This neurophysiological

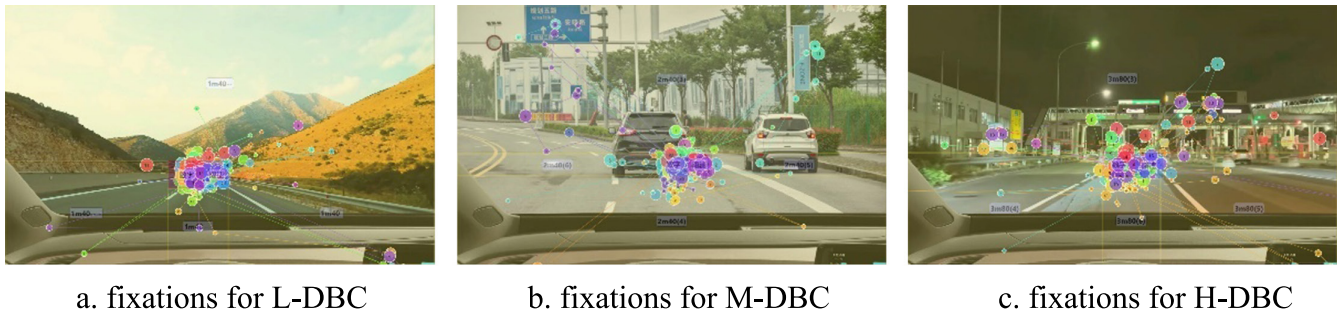


FIGURE 7 | Distribution diagram of the fixation points of the subjects under different DBC conditions.

evidence supports the inference of this study: under H-DBC conditions, drivers do experience increased cognitive load, and changes in their pupil diameter can be regarded as the result of combined cognitive and physiological influences.

This finding highlights one limitation of the current study: The reliance solely on subjective evaluations to assess BDC may lead to biased conclusions. Therefore, future research should aim to establish a more objective and comprehensive approach for evaluating DBC, thereby ensuring a more robust understanding of its impact on cognitive load.

6.2.2 | Influence of DBC on Visual Attention

The experimental results indicate that the GDB-AOI under L-DBC conditions is significantly shorter than that under M-DBC and H-DBC conditions, while no significant difference is observed between M-DBC and H-DBC conditions. This suggests that increasing DBC leads to greater allocation of drivers' attention to the driving environment. Under H-DBC, when drivers' attention is predominantly focused on the driving background and disengagement from complex environmental information becomes challenging, critical AR-HUD display information may be overlooked or not promptly noticed, thereby compromising the effectiveness of AR-HUD's alert function.

As illustrated in Figure 7, the distribution of participants' fixation points under varying driving background complexities reveals distinct patterns. Under L-DBC conditions, fixation points are primarily concentrated in the "current speed" AOI and the "speed limit" AOI, while fewer fixations occur in the background AOI. In contrast, under H-DBC conditions, participants' fixation points are more frequently distributed across the background AOI. These findings align with the research of Yuan [50], which demonstrates that drivers are differentially attracted to external scenes depending on driving scenarios, leading to variations in road distraction duration as their gaze shifts from the road to other areas.

According to the work of Tivesten [51], drivers allocate more time to viewing the road and less to distant perspectives. This study corroborates and extends these conclusions, demonstrating that H-DBC increases drivers' focus on the immediate road and environmental conditions, making it more difficult to attend to AR-HUD display information. These insights offer a novel perspective for AR-HUD design: tailoring AR-HUD displays to account for varying driving background complexities

can significantly enhance visual cognitive efficiency and improve the safety of AR-HUD applications. The AR-HUD interface design can be further refined in the future by analyzing the impact of different matters in the DBC on gaze behavior and visual attention allocation.

Furthermore, the distribution of visual attention while driving is influenced not only by DBC but also by the driving task [52]. Excessive focus on AR-HUD information by drivers can prolong road distraction time and increase the risk of traffic accidents [53]. In future studies addressing AR-HUD display design under varying driving background complexities, it is essential to consider a broader range of influencing factors and conduct more comprehensive investigations of visual cognitive efficiency and driving safety.

6.3 | Limitations and Future Work

This investigation has several methodological constraints that should be acknowledged. First, the selection of DBC levels relied primarily on subjective evaluation. Although standardized assessment metrics were employed, this approach may not fully capture the multidimensional nature of environmental complexity. Second, the experimental design utilized static driving backgrounds rather than dynamic, real-world scenarios with continuously evolving visual stimuli. This simplification limits the ecological validity of the findings. Third, the analysis focused exclusively on drivers' physiological responses to speed displays and DBC variations under controlled conditions. Psychological state fluctuations and performance metrics in operational contexts (e.g., steering inputs, braking latency, and velocity modulation) were not considered. Fourth, cognitive load assessment depended solely on pupillometric measures. As noted by Eckstein et al. [54], multimodal approaches combining pupil dynamics with blink rate analysis provide more robust cognitive state evaluation through complementary psychophysiological indicators.

In future research, we will further explore the following directions to enhance the application value of this study. First, while the current experiments employed static scenarios, subsequent work should validate these findings in dynamic driving environments. This could involve integrating in-vehicle AR-HUD systems with naturalistic driving data to assess the real-time adaptability of opacity designs. Second, given that actual driving involves multitasking (e.g., navigation guidance, emergency braking, and lane maintenance), opacity requirements may

vary substantially across different task types. Future studies should employ a multitask experimental paradigm to systematically examine the interplay between opacity parameters, task categories, and priority levels, thereby optimizing safety across diverse scenarios. Finally, the assessment of DBC warrants refinement. We will test H-DBC scenes under matched luminance levels (e.g., simulated nighttime scenes with adjusted brightness equivalent to daytime) and explicitly incorporate ambient illumination (low/high) into the experimental design to quantify its interaction with DBC and opacity. A hybrid evaluation framework—prioritizing objective metrics while incorporating subjective ratings as supplementary measures—would enable more comprehensive complexity characterization.

7 | Conclusion

This study examines the impact of DBC and AR-HUD opacity on visual cognitive efficiency and visual perceptual characteristics. Initially, a subjective evaluation was conducted to assess DBC, which led to the classification into three levels: low, medium, and high. Subsequently, the complexity of color variety, edge density, and texture features were selected for the objective assessment of the chosen L-DBC, M-DBC, and H-DBC images. Finally, an eye-tracking device was employed to record RTs and visual behaviors. The effects of varying DBCs and text opacity on participants' RTs, APD, and gaze duration were systematically analyzed.

The results reveal significant interactions between DBC and text opacity levels, offering key insights for AR-HUD text opacity optimization. To optimize visual identification performance, AR-HUD transparency should be set at a minimum of 0.6. When AR-HUD transparency is set at 0.7, visual identification efficiency remains relatively stable, and faster RTs are achieved, regardless of whether the DBC is high or low. When opacity levels less than 0.7, the greater the DBC, the longer the response time for the same opacity. Through eye-tracking data, we explored the causes of these phenomena. Complex driving backgrounds increase pupil diameter and prolong gaze durations on background areas, thereby increasing cognitive load and shifting attention away from AR-HUD imagery.

These findings highlight the need to integrate DBC into AR-HUD design to improve driving performance and safety. The different behavioral performances corresponding to varying DBCs indicate that considering DBC is essential in AR-HUD related research. This study offers cognitive ergonomics theoretical support for human–computer interaction design in intelligent vehicles. Future research should explore additional factors affecting visual cognitive efficiency and driving safety, with a focus on optimizing the opacity and presentation of AR-HUD information. This will enhance driver situational awareness and cognitive efficiency in complex driving environments, thereby further advancing AR-HUD system development.

Ethics Statement

After consideration by the Institutional Review Board of this institution, the experimental design and protocol of the study were found to be scientifically sound, fair and impartial, and did not cause harm or

risk to the subjects, the recruitment of subjects was found to be based on the principle of voluntary and informed consent and protection of the rights and interests of the participants as well as their privacy, and the content of the study was found to be free from conflict of interest as well as violation of moral and ethical principles and legal prohibitions, and it complied with the ethical standards set forth in the Declaration of Helsinki. The Institutional Review Board agreed that the work on this project was proceeding as planned.

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